MATH329 · OPERATIONS RESEARCH

Lecture 5 Big-M Method

Midterm summary · Chapter 3 · Artificial variable techniques

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What's inside

01 Why do we need Big-M?

02 The penalty M

03 The steps

04 Example 1: min

05 Example 2: max with an = constraint

06 Example 3: min with three constraints

07 Remarks from the slides

08 Common mistakes

01 Why do we need Big-M?

Simplex starts from the slack variables, because they give the identity matrix directly. That happens when every constraint is `≤`. A `≥` constraint gets a surplus with a minus sign, and an `=` constraint gets nothing, so there is no identity matrix to start from.

> LPP in which all the constraints are (≤) with nonnegative right-hand sides offer a convenient all-slack starting basic feasible solution. Models involving (=) and/or (≥) constraints do not.
> The procedure for starting "ill-behaved" LPP with (=) and (≥) constraints is to use artificial variables that play the role of slacks at the first iteration, and then dispose of them legitimately at a later iteration.

The fix: add a temporary variable called an **Artificial variable**, written `R`, to every `≥` or `=` constraint. It completes the identity matrix, so we can start.

Constraint	What we add
`≤`	A slack `+s` only
`≥`	A surplus `−s` and an artificial `+R`
`=`	An artificial `+R` only

02 The penalty M

`R` is not part of the original problem, so it must leave and equal zero in the final solution. To force it out, we give it a huge coefficient `M` in the objective (a very large positive

number):

The artificial variables are assigned a very high penalty in the objective function, thus forcing them (eventually) to equal zero in the optimum solution:
= –M in a maximization problem
= M in a minimization problem
where M is a very large positive number.

Problem	Coefficient of R in the objective	Why
Max	`−M`	Any positive `R` lowers `z` a lot, so Max drives it out.
Min	`+M`	Any positive `R` raises `z` a lot, so Min drives it out.

03 The steps

#	Step
1	Write the standard form: a slack for `≤` and a surplus for `≥`.
2	Add `R` to every `≥` or `=` constraint, and add `−MR` (max) or `+MR` (min) to the objective.
3	Write the first tableau. The `Z` row comes from `z − (objective) = 0`, so every coefficient flips sign: `R` has `M` under it in max and `−M` in min.
4	Fix the `Z` row: `R` is basic, so its coefficient in the `Z` row must become zero (a requirement to start simplex). Use the formula below.
5	Continue simplex as usual with the rules of Lecture 4.

(New z-row = Old z-row + (M·R1-row + M·R2-row)) min

(New z-row = Old z-row − (M·R1-row + M·R2-row)) max

How to compare values with M

`M` is bigger than any number, so compare the coefficient of `M` first. For example `−2 + 4M` is larger than `−1 + 2M` because `4M > 2M`. If the `M` parts are equal, compare the plain number. For min the most positive enters, and for max the most negative enters, as in Lecture 4.

04 Example 1: min

Minimize $z = 2x_1 + x_2$
subject to $3x_1 + x_2 \geq 9$
$x_1 + x_2 \geq 6$
$x_1, x_2 \geq 0$

1 Standard form + artificial

Minimize $z = 2x_1 + x_2 + 0s_1 + 0s_2 + MR_1 + MR_2$
subject to $3x_1 + x_2 - s_1 + R_1 = 9$
$x_1 + x_2 - s_2 + R_2 = 6$
$x_1, x_2, s_1, s_2, R_1, R_2 \geq 0$

2 The tableau before the fix

Basic	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
Z	−2	−1	0	0	−M	−M	0
R_1	3	1	−1	0	1	0	9
R_2	1	1	0	−1	0	1	6

R_1 and R_2 have $-M$ under them, and they must be zero. It is a min problem, so we add:

```
New Z = Old Z + M(R₁ row) + M(R₂ row)
```

```
x₁:  −2 + M(3) + M(1) = −2 + 4M      Sol.:  0 + 9M + 6M = 15M
```

3 The iterations

Basic	x_1	x_2	s_1	s_2	R_1	R_2	Sol.
Z	$4M - 2$	$2M - 1$	$-M$	$-M$	0	0	$15M$
R_1	**3**	1	-1	0	1	0	9
R_2	1	1	0	-1	0	1	6
Z	0	$\frac{2M}{3} - \frac{1}{3}$	$\frac{M}{3} - \frac{2}{3}$	$-M$	$\frac{2}{3} - \frac{4M}{3}$	0	$3M + 6$
x_1	1	$\frac{1}{3}$	$-\frac{1}{3}$	0	$\frac{1}{3}$	0	3
R_2	0	$\frac{2}{3}$	$\frac{1}{3}$	-1	$-\frac{1}{3}$	1	3
Z	0	0	$-\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2} - M$	$\frac{1}{2} - M$	$\frac{15}{2}$
x_1	1	0	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$
x_2	0	1	$\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{1}{2}$	$\frac{3}{2}$	$\frac{9}{2}$

- Iteration 1: the most positive is $-2 + 4M$ under x_1. The ratios are $\frac{9}{3} = 3$ and $\frac{6}{1} = 6$, so R_1 leaves.
- Iteration 2: the most positive is under x_2. The ratios are $\frac{3}{\frac{1}{3}} = 9$ and $\frac{3}{\frac{2}{3}} = 4.5$, so R_2 leaves.
- Every entry in the Z row is now ≤ 0 (since M is large), so we stop.

$$\text{Min } z = \frac{15}{2} \quad \text{at} \quad x_1 = \frac{3}{2}, \quad x_2 = \frac{9}{2}$$

$$\text{Check:} \quad 2\left(\frac{3}{2}\right) + \frac{9}{2} = 3 + 4.5 = 7.5 \ \checkmark$$

05 Example 2: max with an = constraint

Maximize $z = 2x_1 + 3x_2 - 5x_3$
subject to $x_1 + x_2 + x_3 = 7$
$2x_1 - 5x_2 + x_3 \geq 10$
$x_1, x_2, x_3 \geq 0$

1 Standard form + artificial

The first constraint is `=`, so it takes only R_1. The second is `≥`, so it takes $-s_2$ and R_2.

Maximize $z = 2x_1 + 3x_2 - 5x_3 + 0s_2 - MR_1 - MR_2$
subject to $x_1 + x_2 + x_3 + R_1 = 7$
$2x_1 - 5x_2 + x_3 - s_2 + R_2 = 10$

2 The tableau before the fix

Basic	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
Z	−2	−3	5	0	M	M	0
R_1	1	1	1	0	1	0	7
R_2	2	−5	1	−1	0	1	10

It is a max problem, so we subtract: `New Z = Old Z − M(R1 row) − M(R2 row)`.

3 The iterations

Basic	x_1	x_2	x_3	s_2	R_1	R_2	Sol.
Z	$-3M-2$	$4M-3$	$5-2M$	M	0	0	$-17M$
R_1	1	1	1	0	1	0	7
R_2	**2**	-5	1	-1	0	1	10
Z	0	$-\frac{7M}{2}-8$	$6-\frac{M}{2}$	$-\frac{M}{2}-1$	0	$\frac{3M}{2}+1$	$10-2M$
R_1	0	$\frac{7}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	1	$-\frac{1}{2}$	2
x_1	1	$-\frac{5}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	0	$\frac{1}{2}$	5
Z	0	0	$\frac{50}{7}$	$\frac{1}{7}$	$M+\frac{16}{7}$	$M-\frac{1}{7}$	$\frac{102}{7}$
x_2	0	1	$\frac{1}{7}$	$\frac{1}{7}$	$\frac{2}{7}$	$-\frac{1}{7}$	$\frac{4}{7}$
x_1	1	0	$\frac{6}{7}$	$-\frac{1}{7}$	$\frac{5}{7}$	$\frac{1}{7}$	$\frac{45}{7}$

- Iteration 1: the most negative is $-2-3M$ under x_1. The ratios are $\frac{7}{1}=7$ and $\frac{10}{2}=5$, so R_2 leaves.
- Iteration 2: the most negative is under x_2. The x_1 row has a negative coefficient, so we skip it, and R_1 leaves.

$$\text{Max } z=\frac{102}{7} \quad \text{at} \quad x_1=\frac{45}{7},\ x_2=\frac{4}{7},\ x_3=0$$

06 Example 3: min with three constraints

$$\begin{aligned} &\text{Minimize } z = 4x_1 + 6x_2 \\ &\text{subject to} \quad -2x_1 + 3x_2 = 3 \\ &\qquad\qquad 4x_1 + 5x_2 \geq 10 \\ &\qquad\qquad 4x_1 + 8x_2 \geq 5 \\ &\qquad\qquad x_1, x_2 \geq 0 \end{aligned}$$

$$\begin{aligned} &\text{Minimize } z = 4x_1 + 6x_2 + MR_1 + MR_2 + MR_3 \\ &\text{subject to} \quad -2x_1 + 3x_2 + R_1 = 3 \\ &\qquad\qquad 4x_1 + 5x_2 - s_2 + R_2 = 10 \\ &\qquad\qquad 4x_1 + 8x_2 - s_3 + R_3 = 5 \end{aligned}$$

Basic	x_1	x_2	s_2	s_3	R_1	R_2	R_3	Sol.
Z	$6M - 4$	$16M - 6$	$-M$	$-M$	0	0	0	$18M$
R_1	-2	3	0	0	1	0	0	3
R_2	4	5	-1	0	0	1	0	10
R_3	4	**8**	0	-1	0	0	1	5
Z	$-2M - 1$	0	$-M$	$M - \frac{3}{4}$	0	0	$\frac{3}{4} - 2M$	$8M + \frac{15}{4}$
R_1	$-\frac{7}{2}$	0	0	$\frac{3}{8}$	1	0	$-\frac{3}{8}$	$\frac{9}{8}$
R_2	$\frac{3}{2}$	0	-1	$\frac{5}{8}$	0	1	$-\frac{5}{8}$	$\frac{55}{8}$
x_2	$\frac{1}{2}$	1	0	$-\frac{1}{8}$	0	0	$\frac{1}{8}$	$\frac{5}{8}$
Z	$\frac{22M}{3} - 8$	0	$-M$	0	$2 - \frac{8M}{3}$	0	$-M$	$5M + 6$
s_3	$-\frac{28}{3}$	0	0	1	$\frac{8}{3}$	0	-1	3
R_2	$\frac{22}{3}$	0	-1	0	$-\frac{5}{3}$	1	0	5
x_2	$-\frac{2}{3}$	1	0	0	$\frac{1}{3}$	0	0	1
Z	0	0	$-\frac{12}{11}$	0	$\frac{2}{11} - M$	$\frac{12}{11} - M$	$-M$	$\frac{126}{11}$
s_3	0	0	$-\frac{14}{11}$	1	$\frac{6}{11}$	$\frac{14}{11}$	-1	$\frac{103}{11}$
x_1	1	0	$-\frac{3}{22}$	0	$-\frac{5}{22}$	$\frac{3}{22}$	0	$\frac{15}{22}$
x_2	0	1	$-\frac{1}{11}$	0	$\frac{2}{11}$	$\frac{1}{11}$	0	$\frac{16}{11}$

- Iteration 1: x_2 enters ($-6 + 16M$). The ratios are 1, 2 and $\frac{5}{8}$, so R_3 leaves.
- Iteration 2: the only positive entry is $M - \frac{3}{4}$ under s_3, so the surplus enters. The ratios are 3 and 11, so R_1 leaves.

- Iteration 3: x_1 enters. The only row with a positive coefficient is R_2, so it leaves.

$$\text{Min } z = \frac{126}{11} \quad \text{at} \quad x_1 = \frac{15}{22}, \quad x_2 = \frac{16}{11}$$

$$\text{Check: } -2\left(\frac{15}{22}\right) + 3\left(\frac{16}{11}\right) = -\frac{15}{11} + \frac{48}{11} = 3 \checkmark$$

Note on the slides (checked by calculation): in Example 1 the last tableau has $\frac{1}{2} - M$ under R_1 and R_2 (the slide shows $\frac{1}{2} - \frac{M}{2}$). In Example 3: the second tableau has $\text{Sol.} = \frac{15}{4} + 8M$ (the slide shows $\frac{5}{4} + 8M$), the third tableau has $2 - \frac{8M}{3}$ under R_1 (the slide shows $2 + \frac{8M}{3}$), and the last tableau has $-\frac{12}{11}$ under s_2 (the slide shows $-\frac{6}{11}$) and $x_1 = \frac{15}{22}$ (the table shows $\frac{15}{11}$; the text below it is correct). The final answer does not change.

07 Remarks from the slides

1. If in any iteration, there is a tie for **entering** variable between an artificial variable and other variable (decision, surplus or slack), we must prefer the **non-artificial** variable to enter the basis.
2. If in any iteration, there is a tie for **leaving** variable between an artificial variable and other variable, we must prefer the **artificial** variable to leave the basis.
3. If in the final optimal tableau, an artificial variable is present in the basis at a non-zero level, this means our original problem has **no feasible solution**.

#	In short
1	A tie for entering: prefer the ordinary variable, not R.
2	A tie for leaving: prefer R to leave.
3	If R stays in the final tableau with a nonzero value, the problem is **infeasible** (it has no solution).

08 Common mistakes

Mistake	Correct
Using `+M` in max	Max takes `−MR`, min takes `+MR`.
Adding `R` to a `≤` constraint	`R` goes only with `≥` and `=`.
Forgetting the surplus with `≥`	`≥` takes both `−s` and `+R`.
Starting simplex without fixing the `Z` row	The coefficient of `R` in the `Z` row must become zero first.
Treating `M` as an ordinary number	The coefficient of `M` decides the comparison.